

CHAPTER 23 SUMMARY

Quantization

(Section 23.1) **Quantization** means that a physical quantity has a smallest basic unit, an indivisible **quantum**. Early work in chemistry gave evidence for quantization of mass at the atomic level, while Thomson and Millikan added quantization of charge.

Electric charge is quantized in base units of $\pm e$.

Blackbody Radiation and Planck's Constant

(Section 23.2) **Blackbody radiation** is electromagnetic radiation emitted by hot objects. A variation of **Wien's law** relates the median wavelength λ_{med} and temperature T , where λ_{med} is the wavelength above and below which a blackbody emits half its power. The **Stefan-Boltzmann law** shows that the total power emitted by a blackbody increases rapidly with temperature. Planck was able to describe blackbody radiation, but his description introduced the new constant h and implied that the energies of atomic oscillators are quantized.

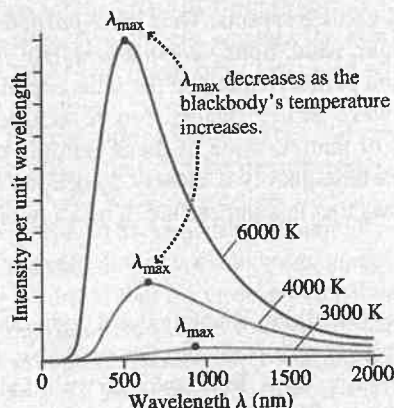
Wien's law: $\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$

Median wavelength: $\lambda_{\text{med}} T = 4.107 \times 10^{-3} \text{ m} \cdot \text{K}$

Stefan-Boltzmann law: $P = \epsilon \sigma A T^4$

Planck's energy quantum: $E = hf$

Planck's constant: $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$



Photons

(Section 23.3) In the **photoelectric effect**, light ejects electrons from a metal. The **threshold frequency** is the minimum needed for emission of photoelectrons from a particular metal used. Einstein explained the photoelectric effect by introducing **quantization of light energy**, suggesting that light consists of particle-like energy bundles now called **photons**. In the **Compton effect**, photons interact with electrons just like particles would, recoiling with lower energy and longer wavelength. Photons have zero mass and momentum inversely proportional to wavelength. **Pair production** converts photon energy into the rest energy of a particle-antiparticle pair. The reverse process is **pair annihilation**: When a particle and antiparticle meet, they disappear and are replaced by two photons.

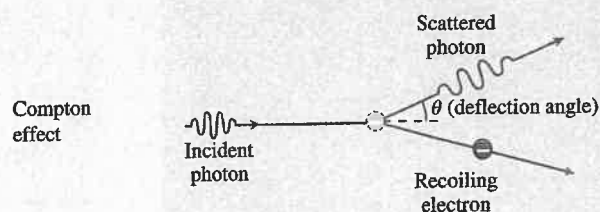
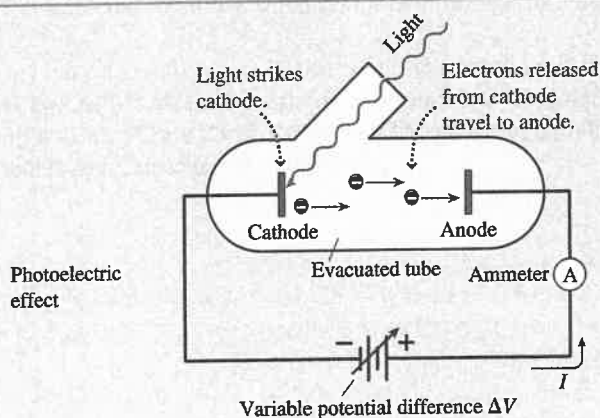
Quantized photon energy: $E = hf = \frac{hc}{\lambda}$

Einstein's equation: $eV_0 = K_{\text{max}} = hf - W_0$

The **work function** W_0 is the energy required to eject an electron from a particular metal.

Photon momentum, energy, and wavelength: $p = \frac{E}{c} = \frac{h}{\lambda}$

Compton effect: $\Delta\lambda = \lambda - \lambda_0 = \frac{h}{mc}(1 - \cos\theta)$

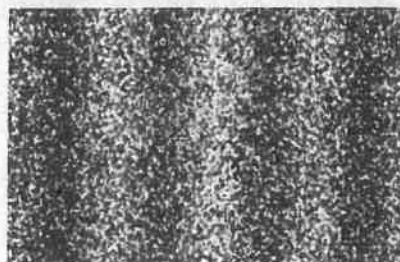


Wave-Particle Duality

(Section 23.4) Wave-particle duality describes the fact that light exhibits both particle and wave behaviors. De Broglie suggested that wave-particle duality applies to matter, and electron diffraction experiments show electrons acting like waves. The **Heisenberg uncertainty principle** limits the precision with which a particle's momentum and position can be measured simultaneously.

de Broglie wavelength: $\lambda = \frac{h}{p}$

Heisenberg uncertainty principle: $\Delta p_x \Delta x \geq \frac{h}{4\pi}$



NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled B10 are of biological or medical interest.

Conceptual Questions

1. Why was helium first observed on the Sun?
2. Which has a higher temperature—a red-hot object or a white-hot one?
3. What is the “ultraviolet catastrophe”? In what sense is it a “catastrophe”?
4. A red laser and a blue laser emit the same number of photons each second. Which laser emits more power?
5. A red laser and a blue laser have the same power. Which (if either) emits more photons per second?
6. When a photon scatters from an electron initially at rest, why does the photon's wavelength increase?
7. What is a metal's “work function”? Why is it different for different metals?
8. Why is there a threshold frequency for photons in the photoelectric effect?
9. A particular metal ejects photoelectrons when illuminated with green light but not yellow. Do you expect that photoelectrons will be ejected from this metal with (a) orange light; (b) blue light?
10. From the standpoint of classical physics, why is it surprising that electrons are ejected almost immediately in a photoelectric experiment? How does quantum physics resolve this quandary?
11. An electron is traveling at a high but nonrelativistic speed. If its speed doubles, what happens to its de Broglie wavelength?
12. A rock is dropped from a tall building. Ignoring the effects of air resistance, how does its de Broglie wavelength vary as it falls?
13. It has been estimated that the proton's lifetime is at least 10^{36} years. Does that suggest we can know the proton's rest energy precisely, or only approximately?
14. Do baseball batters need to worry about the Heisenberg uncertainty principle when they swing at the ball?
15. Speculate on how the uncertainty principle affects the notion of objectivity in science.
16. Compare the relative advantages of the transmission electron microscope and the scanning electron microscope.

Multiple-Choice Problems

17. How many charge quanta are there in 20 mC? (a) 3.2×10^{16} ; (b) 1.2×10^{17} ; (c) 3.2×10^{17} (d) 1.3×10^{18} .
18. How many hydrogen atoms are there in a 1-g sample of pure hydrogen? (a) 6.0×10^{23} ; (b) 6.0×10^{24} ; (c) 6.0×10^{25} ; (d) 6.0×10^{26} .
19. A blackbody radiates with median wavelength 2200 nm. What's its temperature? (a) 975 K; (b) 1050 K; (c) 1850 K; (d) 3020 K.
20. A blackbody heated to 900°C emits radiation at the rate of 850 W. If the blackbody's temperature is increased to 1200°C, the power becomes (a) 1130 W; (b) 1340 W; (c) 2110 W; (d) 2700 W.
21. The energy of one quantum of 12-GHz microwave radiation is (a) 8.0×10^{-22} J; (b) 4.0×10^{-23} J; (c) 4.0×10^{-24} J; (d) 8.0×10^{-24} J.
22. A 50-mW laser produces 532-nm green light. The number of photons it emits each second is (a) 1.3×10^{17} ; (b) 1.6×10^{17} ; (c) 2.7×10^{17} ; (d) 2.7×10^{18} .
23. Silver's work function is 4.64 eV. The longest wavelength that will eject photoelectrons from silver is (a) 535 nm; (b) 400 nm; (c) 361 nm; (d) 267 nm.
24. What's the energy of a photon with wavelength 590 nm? (a) 1.1 eV; (b) 2.1 eV; (c) 4.2 eV; (d) 5.8 eV.
25. What's the momentum of a photon with wavelength 600 nm? (a) 3.2×10^{-26} kg·m/s; (b) 5.5×10^{-26} kg·m/s; (c) 7.5×10^{-26} kg·m/s; (d) 1.1×10^{-24} kg·m/s.
26. An x-ray photon with wavelength 2.98 pm scatters from a stationary electron, deflecting through a 55° angle. What's the wavelength of the scattered photon? (a) 3.06×10^{-3} nm; (b) 3.57×10^{-3} nm; (c) 4.02×10^{-3} nm; (d) 4.56×10^{-3} nm.
27. If they're all traveling at the same speed, which particle has the shortest de Broglie wavelength? (a) proton; (b) neutron; (c) electron; (d) alpha particle.
28. Electrons in an electron microscope must have de Broglie wavelengths of 0.45 nm. What's the electrons' speed? (a) 5.6 × 10⁵ m/s; (b) 1.6 × 10⁶ m/s; (c) 7.2 × 10⁶ m/s; (d) very close to the speed of light.

Problems

Section 23.1 Quantization

29. ■ How many quanta of charge are present in (a) 1 C; (b) 1 μC ?
30. ■ A toy balloon is filled with helium gas (He) at 1 atm and 20°C. If the volume of the balloon is 0.027 m³, how many atoms does it contain?
31. ■■ You have a piece of solid iron containing 2.50 moles of atoms. (a) What's its mass? (b) Suppose one atom in 10¹² were missing one electron. What would be the net charge on the iron? Would you notice this if you touched the iron?
32. ■■ An antiproton is similar to a proton, but with charge $-e$. What are the charges on the three quarks that form an antiproton?

Section 23.2 Blackbody Radiation and Planck's Constant

33. ■ Radiation from a blackbody has its Wien peak at 1270 nm. What's the blackbody's temperature?
34. ■ A piece of iron is heated to 600°C. (a) What is the median wavelength of its blackbody radiation? (b) What temperature would give a median wavelength of half of what you found in part (a)?
35. ■ Find the surface temperature of a star whose Wien peak is in the near ultraviolet, with $\lambda = 390$ nm.
36. ■ A red giant star has surface temperature 3100 K. What's the median wavelength in its blackbody spectrum? Compare with the Sun (see Making the Connection in Example 23.1).
37. ■■ The tungsten lightbulb filament operates at about 3000 K (not far below tungsten's melting point). (a) What's the median wavelength it emits, and in what region of the spectrum does this lie? (b) If it's a 100-W bulb, what's the filament's surface area?
38. ■ A white dwarf star has surface temperature 20 kK. (a) Where's the Wien peak in this star's blackbody curve? (b) In what part of the electromagnetic spectrum is that peak?
39. ■ In the 1960s astronomers detected blackbody radiation with Wien peak at 1.06 mm, apparently coming from everywhere in space. What's the temperature of the radiation source? (This "cosmic microwave background" radiation was key to understanding the evolution of the universe.)
40. ■■ A blue supergiant star has surface temperature 30 kK and has total power output 100,000 times that of the Sun. (a) Where is the Wien peak of this star's blackbody curve? (b) Why does the star appear blue? (c) Given the Sun's 6.96×10^8 m radius, what's the radius of the supergiant?
41. ■■ An 800-K blackbody emits 450 W of radiation. At what temperature will the radiated power double to 900 W?

Section 23.3 Photons

42. ■ A photoelectric experiment uses 265-nm light and a silver target. Find (a) the maximum kinetic energy of the photoelectrons and (b) the stopping potential.
43. ■■ Find the threshold frequency and wavelength for production of photoelectrons from a lead target.
44. ■■ What's the longest wavelength that will produce photoelectrons from iron? In what part of the electromagnetic spectrum is this?
45. ■■ (a) What are the threshold frequency and wavelength for photons from a silver target? (b) What's the stopping potential for photoelectrons from silver under 135-nm ultraviolet light?
46. ■■ Suppose you want to produce photoelectrons with kinetic energy 1.50 eV from a copper target. What wavelength of electromagnetic radiation should you use?
47. ■■ A sodium target is illuminated with 442-nm violet light. Find (a) the stopping potential for the photoelectrons produced and (b) the photoelectrons' maximum speed.

48. ■■ Of the elements in Table 23.1, which can produce photoelectrons when illuminated with visible light?
49. ■■ Photoelectrons from a metal target have a 1.20-V stopping potential when the target is illuminated with 340-nm ultraviolet radiation. Find (a) the stopping potential for the same target under 260-nm radiation and (b) the work function for this metal.
50. ■■ A 1.0-mW laser with a 405-nm wavelength illuminates a sodium target. If 1 in 10⁵ incident photons generates a photoelectron, what's the photocurrent?
51. ■■■ In an experiment to measure Planck's constant, a metal target is illuminated with ultraviolet light. When the wavelength is 300 nm, the measured stopping potential is 1.10 V. When the wavelength is changed to 200 nm, the stopping potential becomes 3.06 V. (a) What value does this experiment yield for Planck's constant? What's the percent error from the accepted value of h ? (b) Using the experimental value h , find the threshold frequency and work function for this metal.
52. ■ An x-ray photon with wavelength 1.94 pm scatters from a stationary electron, deflecting through an angle of 105°. What's the scattered photon's wavelength?
53. ■ Find the energy of one quantum of microwave radiation with frequency (a) 1 GHz and (b) 300 GHz.
54. ■■ When perceiving 630-nm red light, your unaided eye can barely detect light at a threshold power around 2.5×10^{-15} W. At what rate are photons entering your eye at this level?
55. ■■ A semiconductor laser produces a continuous 10.0-W beam with wavelength 2.4 μm . Find (a) the energy of each photon and (b) the number of photons emitted each second.
56. ■■ (a) Find the wavelength of a 4.1-eV photon. In what part of the electromagnetic spectrum is this? Repeat for photons with energies (b) 2.3 eV and (c) 0.69 eV.
57. ■■ While sitting in direct sunlight, you absorb UVB radiation with average wavelength 310 nm. (a) Find the energy of an average UVB photon, and compare with an average visible photon ($\lambda = 550$ nm). (b) How many photons are in 1 J of energy at each wavelength?
58. ■■ A medical x-ray beam has wavelength 1.26 nm. (a) What's the energy of one photon in this beam? (b) How many photons are in a typical chest x-ray dose that delivers 0.021 J of energy?
59. ■■■ Example 23.3 showed that the Sun emits energy at the rate of about 3.9×10^{26} W. (a) Using the astronomical data in Appendix E, estimate the fraction of this energy that falls on Earth. (b) With the Sun directly overhead, what's the rate at which energy strikes each square meter of Earth's surface? (Ignore reflection and absorption by the atmosphere.) (c) Discuss the implications of your answer in part (b) for solar energy use.
60. ■■ A 639 keV x-ray photon scatters from an electron at rest, deflecting through 105°. What is the scattered photon's energy?
61. ■■ You're studying Compton scattering using x rays with wavelength equal to the Compton wavelength. At what scattering angle will you observe x rays with a wavelength twice that of the incident x rays?
62. ■■ A photon with a wavelength 4.50 pm scatters from an electron initially at rest. Find (a) the maximum kinetic energy for the scattered electron and (b) the wavelength of the scattered photon when the scattered electron has its maximum kinetic energy.
63. ■■ X rays with wavelength 5.0 pm scatter from electrons, recoiling with wavelength 8.2 pm. What's the scattering angle?
64. ■■ Compton scattering can occur from atoms as well as from electrons. If photons scatter from atoms of helium (He) gas, what's their maximum wavelength change? Should the photon energies be larger or smaller than those used in Compton scattering by electrons?

65. ■■ Your Compton-effect instrumentation can detect a 5.0% change in the photon's wavelength. What's the maximum initial photon wavelength in this case?
66. ■■ A hydrogen atom initially at rest emits a 434-nm photon. Find (a) the photon's energy and momentum, (b) the momentum of the atom after emitting the photon, and (c) the kinetic energy of the recoiling atom. Compare with the photon's energy.
67. ■■ (a) Find the energy needed to produce a proton-antiproton pair. (b) Find the energies of the two photons produced when a proton and antiproton at rest annihilate.
68. ■■■ In a colliding-beam experiment, an electron and positron traveling at the same speeds but opposite directions collide, producing two photons. If each of the photons has energy equal to three times the electron's rest energy, what were the electron and positron speeds?

Section 23.4 Wave-Particle Duality

69. ■ Find (a) the speed of an electron with de Broglie wavelength 1.0 nm and (b) the de Broglie wavelength of a proton with that speed.
70. ■ The electrons in an electron microscope have speed of 4.0 Mm/s. Find their de Broglie wavelength.
71. ■ An electron microscope calls for electrons with de Broglie wavelength 0.25 nm. What's the electrons' speed?
72. ■■ In an undergraduate version of the Davisson-Germer experiment, electrons are accelerated through several potential differences. For each potential difference, find the corresponding de Broglie wavelength and electron speed: (a) 12 V; (b) 30 V; (c) 60 V.
73. ■■ Find the kinetic energy of each of the following particles, each having de Broglie wavelength 0.30 nm: (a) electrons; (b) neutrons; (c) alpha particles.
74. ■■ Suppose you want to observe neutron diffraction in a crystal with 0.29-nm spacing between lattice planes. (a) Find the momentum and kinetic energy of neutrons having wavelength equal to this spacing. (b) If these neutrons constituted an ideal gas, what would be the gas temperature? Explain why these are called "cold" neutrons.
75. ■■■ What's the de Broglie wavelength of a nitrogen (N_2) molecule in air at room temperature (293 K)? How likely is it that you'll be able to observe diffraction effects in such molecules?
76. ■■■ A 25-g marble passes through a 2.0-cm-wide opening. What should be the marble's speed in order for it to experience a 0.1° diffraction angle through the opening? (Assume first-order single-slit diffraction.)
77. ■■ (a) What's the kinetic energy of an electron having de Broglie wavelength 400 nm? (b) Compare with the energy of a photon having the same wavelength.
78. ■■ If you want to measure the position of a 142-g baseball moving at 25 m/s to within $1.0 \mu\text{m}$, what's the corresponding uncertainty in its speed? Do you think you can really measure the speed with that precision?
79. ■■■ An electron is confined to a box the size of a small atom, 0.10 nm across. (a) What's the uncertainty in the electron's momentum? (b) Suppose the momentum is just equal to the minimum uncertainty value you computed in part (a). What's the electron's energy? What wavelength photon would have the same energy?
80. ■■ The W^+ is an elementary particle with rest energy around 80 GeV, uncertain by about 2.1 GeV. Use Heisenberg's uncertainty principle to estimate this particle's minimum lifetime.

When a particle is moving with a relativistic speed, its de Broglie wavelength is still $\lambda = h/p$, where p is the relativistic momentum. Use this fact in the following problems.

81. ■■ Find the de Broglie wavelength of an electron moving at (a) 0.10c; (b) 0.50c; (c) 0.99c.
82. ■■ In a certain electron microscope, a de Broglie wavelength of 0.015 nm is required. Find the potential difference required to accelerate electrons to the proper speed.
83. ■■■ The Stanford linear accelerator can accelerate electrons to kinetic energies of 50 GeV. What's the de Broglie wavelength of these electrons? Compare with the diameter of a proton, about 2 fm.
84. ■■■ In his double-slit experiment with electrons, Jönsson used electrons with kinetic energy 50 keV. (a) Find the electrons' de Broglie wavelength. (b) The two slits were separated by $2.0 \mu\text{m}$. On a screen 0.350 m from the slits, what was the distance between bright diffraction fringes?

General Problems

85. ■■ (a) Find the Compton wavelength of a proton. (b) Find the energy (in eV) of a gamma ray whose wavelength equals the proton's Compton wavelength.
86. ■■ Electrons in a photoelectric experiment emerge from an aluminum surface with a maximum kinetic energy of 1.3 eV. What is the wavelength of the incident radiation?
87. ■■ The most energetic cosmic rays ever detected are photons with energies approaching 10 J. Find the wavelength of such a photon.
88. ■■ A cosmic ray interacts for a mere 12 fs with its detector. Having just learned about the uncertainty principle, you realize that this interaction time establishes a precision limit on measurement of cosmic-ray energy. Find (a) the minimum uncertainty in that energy and (b) the fractional uncertainty for a 4.5-MeV cosmic ray.
89. ■■ An electron is initially at rest. What will be its kinetic energy after a 0.10-nm x-ray photon scatters from it at 90° to its original direction?
90. ■■■ A photocathode emits electrons with maximum kinetic energy 0.85 eV when illuminated with 430-nm violet light. (a) Will it eject electrons under 633-nm red light? (b) Find the threshold wavelength for this material.
91. B10 ■■■ **Human vision threshold.** (a) Estimate the number of photons per second emitted by a 100-W lightbulb, assuming a photon wavelength in the middle of the visible spectrum, 550 nm. (b) A person can just see this bulb from a distance of 800 m, with the pupil diameter dilated to 7.5 mm. How many photons per second are entering the pupil?

Answers to Chapter Questions

Answer to Chapter-Opening Question

It's quantum tunneling, the ability of particles to pass through a gap that classical physics says they don't have enough energy to overcome.

Answers to GOT IT? Questions

Section 23.1 (a) $H^+ > (c) He^{+2} > (b) He^+ > (d) Na^+$

Section 23.3 (f) gamma ray $> (d)$ x-ray $> (b)$ ultraviolet $> (a)$ visible $> (e)$ infrared $> (c)$ radio

Section 23.4 (b) decreases